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Method for starting up a wind turbine

Abstract

A method for starting up a wind turbine for generating electrical power is provided, wherein ice accretion is on at least one blade of a rotor of the wind turbine and wherein the at least one blade has a variable pitch angle, the method including measuring the rotational speed of the electromechanical transducer, calculating a pitch angle adjustment with the goal of maximizing the rotational speed, modifying the pitch angle in a positive or negative step according to the pitch angle adjustment, iterate the preceding steps.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2008/0317583	12/2007	Grabau	N/A	N/A
2013/0106107	12/2012	Spruce	290/44	F03D 7/0224
2014/0091572	12/2013	Jepsen	N/A	N/A
2017/0058871	12/2016	Movsichoff et al.	N/A	N/A
2017/0226990	12/2016	Arlaban Gabeiras et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2508316	12/2011	CA	N/A
101124402	12/2009	CN	N/A
103821665	12/2013	CN	N/A
20014238	12/2000	DE	N/A
2559894	12/2012	EP	N/A
2567616	12/2014	RU	N/A
9801340	12/1997	WO	N/A
2004057182	12/2003	WO	N/A

OTHER PUBLICATIONS

PCT International Search Report and Written Opinion of International Searching Authority mailed Feb. 21, 2023 corresponding to PCT International Application No. PCT/EP2022/083602 filed Nov. 29, 2022. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a national stage of PCT Application No. PCT/EP2022/083602, having a filing date of Nov. 29, 2022, which claims priority to EP Application Serial No. 21383153.0, having a filing date of Dec. 17, 2021, the entire contents both of which are hereby incorporated by reference.
FIELD OF TECHNOLOGY
(2) The following relates to the technical field of starting up a wind turbine with ice accretion on a rotor blade. In particular, the following relates to a method for starting up a wind turbine. Further, the following relates to a wind turbine control system and to a wind turbine which are all adapted for executing the method for starting up a wind turbine.
BACKGROUND
(3) Ice accretion or ice formation on the rotor blades remarkably reduces the aerodynamic performance of wind turbines. The shapes of the aerodynamic profiles are modified due to the ice accreted on them and, as a result, their aerodynamic behavior is severely altered. Furthermore, the wind turbine's operation can be threatened due to the aerodynamic stall phenomena.
(4) By now, the main objective of the operation with ice control functionality is to avoid stop of the wind turbine during icing conditions and maximize its power production. To that end, adaptive pitch and torque functionalities can be implemented in order to try to avoid the wind turbine from suffering aerodynamic stall during icing periods.
(5) There are several methods for the operation with ice or the ice removal from the wind turbine blades.
(6) CA 2508316 C discloses a generator which is used as a motor to remove ice from the blades.
(7) CN 103821665 A, DE 20014238 U1, WO 1998001340 A1 disclose removal of ice using a de-icing system.
(8) CN 101124402 B discloses ice prevention in offshore wind turbines using sea water.
(9) RU 2567616 C2 discloses operation in ice situations based on stopping the wind turbine.
(10) US20170058871 A1 discloses operation in ice situations based on yawing the wind turbines.
(11) Normally, control strategies designed for the operation with ice are focused on maximizing the power production of the turbines during icing conditions. However, these strategies might result useless if the wind turbine is unable to complete the start-up procedure due to the ice accretion.
(12) Therefore, there may be a need for improving the start-up procedure of wind turbines in cold climates.
SUMMARY
(13) An aspect relates to a method for starting up a wind turbine for generating electrical power, wherein ice accretion is on at least one blade of a rotor of the wind turbine and wherein the at least one blade has a variable pitch angle, the method comprising measuring the rotational speed of the electromechanical transducer, calculating a pitch angle adjustment with the goal of maximizing the rotational speed, modifying the pitch angle in a positive or negative step according to the pitch angle adjustment, iterate the preceding steps.
(14) The described method is based on the idea to provide an adaptive pitch strategy for a wind turbine start-up procedure in cold climates. The described method serves to improve the start-up capability of wind turbines affected by ice accretion on the blades.
(15) The pitch angle is modified in positive or negative steps according to the temporal development of the rotational speed and with the objective of maximizing it.
(16) The proposed adaptive pitch strategy improves the start-up process of the wind turbines during icing periods. As the ice accretes on the blades, the aerodynamic profile is changed and the optimal

pitch angle changes to an unknown value. Even due to this worsened and unknown aerodynamic behavior of the blades, the proposed method enables the wind turbine to accelerate and reach the rotational speed necessary to connect the wind turbine to the grid.

(17) The optimization algorithm utilizes the rotational speed as an input. This allows operation even when the wind turbine is not connected to the grid. In that case there is no electrical power to measure.

(18) The described method may have the advantage of an Annual Energy Production (AEP) gain in cold climate sites. In embodiments, the method might enable the wind turbine to start-up and connect to the grid even with ice accretion on the blades. Without this functionality, the wind turbine might suffer important energy production losses.

(19) Further, the described method does not require to install or calibrate any sensor or de-icing system, which is usually not included in a wind turbine. In embodiments, the method is robust, cost competitive and easy to implement.

(20) According to an embodiment of the invention, the pitch angle adjustment is calculated with a cost function, the cost function receiving the rotational speed of the electromechanical transducer as an input with the goal of maximizing the rotational speed. Having ice accretion on the blades, the pitch angle behavior is dynamic and not static anymore. Such situation can be covered well by a cost function. The cost function aims at maximizing the rotational speed by comparing it before and after each adjustment to the pitch angle. A cost function may be designed by a numerical optimization procedure for example. Simulations can be performed to reach values for implementing a cost function. Such cost function may be a scalar cost function.

(21) According to an embodiment of the invention as a first step, ice accretion is detected. Such detection may be achieved by identifying non-plausible or not expected sensor readings, the wind turbine not producing enough energy, the startup procedure not working in normal parameters and/or by weather information.

(22) According to an embodiment of the invention as a last step, the method is terminated after sufficient rotational speed is measured to connect the electromechanical transducer to the grid, a predetermined time limit is reached without being able to start-up the wind turbine and/or no ice is detected.

(23) According to an embodiment of the invention, the measured rotational speed of the electromechanical transducer is filtered with a low pass and/or a band pass. Filters can be implemented as first order or second order. Filtering may enhance robustness of the method.

(24) According to an embodiment of the invention, a first pitch angle adjustment is a pitch step towards feather. Feathering means that the blade pitch is increased to the point that the chord line of the blade is approximately parallel to the on-coming airflow. This approach minimizes the unwanted effect of stalling the wind turbine.

(25) According to an embodiment of the invention, if the pitch angle adjustment towards feather does not lead to an increase of the speed of the electromechanical transducer a pitch angle adjustment towards fine is introduced. The next or second iteration step could be directed to the opposite direction i.e., towards fine. Such approach may deliver better information of the ice situation and could lead to the optimized pitch angle.

(26) According to an embodiment of the invention, amplitudes of the pitch angle adjustments are adaptive with respect to the number of iterations. As the optimal pitch angle for the current ice accretion situation should be closer with an increasing number of iterations the amplitude of the adjustments may be lowered. Further, the pitch movement, according to each adaptation or iteration, may have a maximum and a minimum. This may prevent that the algorithm runs too far in a wrong direction. Adaptation of the amplitude of pitch angle adjustments may help to better adapt to the unknown ice conditions.

(27) According to embodiments of the invention, there is provided a wind turbine control system for controlling a wind turbine having an electromechanical transducer. The control system

comprises a computation unit adapted for executing the method as described above, an input adapted for receiving a rotational speed of the electromechanical transducer, and an output adapted for outputting a pitch angle adjustment. The same advantages and modifications as described above apply.
(28) According to an embodiment of the invention, the computation unit comprises a low pass and/or a band pass filter adapted for filtering the measured rotational speed of the electromechanical transducer. The filter can be implemented as first order or second order. Implementing a filtering stage allows for more reliable and sturdy data handling.
(29) According to a further aspect of embodiments of the invention, there is provided a wind turbine for generating electrical power. The provided wind turbine comprises a tower, a rotor, which is arranged at a top portion of the tower, and which comprises at least one blade, wherein the at least one blade has a variable pitch angle, an electromechanical transducer which is mechanically coupled with the rotor, and a wind turbine control system as described above. The same advantages and modifications as described above apply.
(30) According to an embodiment of the invention, the wind turbine comprises a rotational speed sensor adapted for measuring the rotational speed of the electromechanical transducer and/or a pitch sensor adapted for measuring a pitch angle of the at least one blade. According to the described wind turbine, it is not required to install or calibrate any further sensors or a de-icing system, which is usually not included in a wind turbine. Thus, the improvement is being robust, cost competitive and easy to implement.
(31) It has to be noted that embodiments of the invention have been described with reference to different subject matters. Some embodiments have been described with reference to apparatus type claims whereas other embodiments have been described with reference to method type claims. However, a person skilled in the conventional will gather from the above and the following description that, unless other notified, in addition to any combination of features belonging to one type of subject matter also any combination between features relating to different subject matters, in particular between features of the apparatus type claims and features of the method type claims is considered as to be disclosed with this document.
(32) The aspects defined above and further aspects of the present invention are apparent from the examples of embodiment to be described hereinafter and are explained with reference to the examples of embodiment. The invention will be described in more detail hereinafter with reference to examples of embodiment but to which the invention is not limited.

Description

BRIEF DESCRIPTION
(1) Some of the embodiments will be described in detail, with reference to the following figures, wherein like designations denote like members, wherein:
(2) FIG. **1** shows a wind turbine according to an embodiment of the present invention;
(3) FIG. **2** shows a schematic illustration of a control system of the wind turbine;
(4) FIG. **3** shows a flow diagram of a method for starting up a wind turbine having ice accretion on its blade; and
(5) FIG. **4** shows diagrams of the pitch, the generator speed and the cost according to an implementation of the method depicted in FIG. **3**.
DETAILED DESCRIPTION
(6) FIG. **1** shows a wind turbine **100** according to an embodiment of the invention. The wind turbine **100** comprises a tower **120**, which is mounted on a non-depicted fundament. On top of the tower **120** there is arranged a nacelle **122**. In between the tower **120** and the nacelle **122** there is provided a yaw angle adjustment device **121**, which is capable of rotating the nacelle **122** around a

non-depicted vertical axis, which is aligned with the longitudinal extension of the tower **120**. By controlling the yaw angle adjustment device **121** in an appropriate manner it can be made sure that during a normal operation of the wind turbine **100** the nacelle **122** is always properly aligned with the current wind direction. However, the yaw angle adjustment device **121** can also be used to adjust the yaw angle to a position, wherein the nacelle **122** is intentionally not perfectly aligned with the current wind direction.

(7) The wind turbine **100** further comprises a rotor **110** having three blades **114**. In the perspective of FIG. **1** only two blades **114** are visible. The rotor **110** is rotatable around a rotational axis **110***a*. The blades **114**, which are mounted at a hub **112**, extend radially with respect to the rotational axis **110***a*.

(8) In between the hub **112** and a blade **114** there is respectively provided a blade adjustment device **116** in order to adjust the blade pitch angle of each blade **114** by rotating the respective blade **114** around a non-depicted axis being aligned substantially parallel with the longitudinal extension of the blade **114**. By controlling the blade adjustment device **116** the blade pitch angle of the respective blade **114** can be adjusted in such a manner that at least when the wind is not so strong a maximum wind power can be retrieved from the available wind power. However, the blade pitch angle can also be intentionally adjusted to a position, wherein only a reduced wind power can be captured.

(9) As can be seen from FIG. **1**, the rotor **110** is directly coupled with a shaft **125**, which is coupled in a known manner to an electromechanical transducer **140**. The electromechanical transducer is a generator **140**.

(10) Accordingly, the turbine is a direct drive type, wherein the hub is directly connected to the generator **140**, i.e., no gearbox is present. Alternatively, a gearbox may be arranged between the shaft **125** and the generator **140**.

(11) Further, a brake **126** is provided in order to stop the operation of the wind turbine **100** or to reduce the rotational speed of the rotor **110** for instance (a) in case of an emergency, (b) in case of too strong wind conditions, which might harm the wind turbine **100**, and/or (c) in case of an intentional saving of the consumed fatigue life time and/or the fatigue life time consumption rate of at least one structural component of the wind turbine **100**.

(12) In accordance with basic principles of electrical engineering the generator **140** comprises a stator assembly **145** and a rotor assembly **150**. The generator **140** may include an external rotor **150** which is arranged outside the stator **145**.

(13) The stator assembly **145** comprises a plurality of coils for generating electrical current in response to a time alternating magnetic flux. The rotor assembly comprises a plurality of permanent magnets, which are arranged in rows being aligned with a longitudinal axis of the rotor assembly **150**. The permanent magnets may be skewed to reduce torque ripple.

(14) The wind turbine **100** further comprises a control system **153** for operating the wind turbine **100** in a highly efficient manner. Apart from controlling for instance the yaw angle adjustment device **121** the depicted control system **153** is also used for adjusting the blade pitch angle of the rotor blades **114** in an optimized manner.

(15) The control system **153** is connected with a rotational speed sensor **154** which is adapted for measuring the rotational speed of the generator **140**. The control system **153** is further connected with a pitch sensor **155** which is adapted for measuring a pitch angle of the rotor blades **114**. For ease of understanding power lines and signal lines between the control system **153** and the sensors are not depicted. Details of the control system **153** are discussed in conjunction with the following figures.

(16) In cold climates, ice accretion **156** aggregates on the rotor blades **114** which severely reduces the aerodynamic performance of the wind turbine **100**. The shapes of the aerodynamic profiles of the rotor blades **114** are modified due to the ice accreted on them so that their aerodynamic behavior is altered.

(17) Cold climates can be defined as continental climate and polar climate. Regions in a continental climate tend to have warm to cool summers and very cold winters. In the winter, this climate zone can experience snowstorms, strong winds, and very cold temperatures—sometimes falling below −30° C. For regions in a polar climate zone, the temperatures tend to never increase higher than 10° C. even during summer.
(18) FIG. **2** shows a schematic view of the control system **153** which is adapted to execute or calculate an adaptive pitch strategy for a start-up procedure of the wind turbine **100** in cold climates.
(19) The control system **153** includes a computation unit **157** like a microprocessor or the like. The control system **153** or the computation unit **157**, respectively is adapted for executing a method for starting up a wind turbine **100** when ice accretion is on a blade **114** of the rotor **110** of the wind turbine **100**. In embodiments, the method is described in conjunction with the following figures.
(20) The control system **153** is connected with the rotational speed sensor **154** via a first input **158**. The first input **158** is connected with the computation unit **157**. A value or signal provided by the rotational speed sensor **154** is treated as an input by the control system **153** and is utilized for calculations such as the method or algorithm for starting up a wind turbine **100** with ice accretion.
(21) The control system **153** is further connected with a pitch sensor **155** via a second input **159**. The second input **159** is connected with the computation unit **157**. A single pitch sensor **155** can be implemented or one pitch sensor **155** can be provided for each of the rotor blades **114**. The number of second inputs **159** may equal to the number of pitch sensors **155**.
(22) A value or signal provided by the pitch sensor **155** is treated as an input by the control system **153** and is utilized for calculations such as the method or algorithm for starting up a wind turbine **100** with ice accretion.
(23) The control system **153** has at least one output **160** for outputting control or steering values or signals which can among others be used for adjusting the blade pitch angle of the rotor blades **114**. The control signals are derived from calculations based on the input signal or signals.
(24) FIG. **3** shows a flow diagram of a method for starting up a wind turbine **100** having ice accretion on its blade **114** and/or being located in a cold climate. In embodiments, the method can be executed by or implemented in the control system **153** as described before.
(25) In a first step **300**, the rotational speed of the electromechanical transducer here in form of the generator **140** is measured. The measured rotational speed may be filtered for example with a low or band pass before further processing it.
(26) In some embodiments, as a primary step or as a starting condition, ice accretion is detected. Alternatively, the adaptive pitch strategy of the described method can be set as default for a wind turbine **100** start-up procedure in cold climates.
(27) In a second step **310**, a pitch angle adjustment is calculated with the goal of maximizing the rotational speed of the wind turbine **100**. The rotational speed of the wind turbine **100** corresponds to the measured rotational speed of the generator **140**. With the goal of maximizing the rotational speed of the wind turbine **100** a quick start of the wind turbine **100** can be achieved as the generator needs a certain rotational speed for connecting it to the grid.
(28) The calculation of the pitch angle adjustment may be achieved by a cost function. Such cost function receives the rotational speed of the generator **140** as an input with the goal of maximizing the rotational speed by maximizing the cost. The cost function is explained in conjunction with FIG. **4**.
(29) In a third step **320**, the pitch angle is modified in a positive or negative step or amount according to the calculated pitch angle adjustment.
(30) The preceding steps are iterated in a loop so that effect of the calculated pitch angle adjustment can be evaluated in the calculation step. Depending on the change in rotational speed the pitch angle adjustment of the next iteration will have the same or opposite sign. The same sign may be applied when the rotational speed had increased while the opposite may be applied when the

rotational speed had decreased.
(31) As an optional last step or as a termination condition, the method is terminated after sufficient rotational speed is measured to connect the generator **140** to the grid, a predetermined time limit is reached without being able to start-up the wind turbine **100** and/or no ice is detected, e.g., when implausible sensor readings have stopped.
(32) FIG. **4** shows a diagram **410** of the pitch, a diagram **420** of the generator speed and a diagram **430** of the cost according to an implementation of the method depicted in FIG. **3**. All diagrams are shown over time in seconds.
(33) The described optimization algorithm or method cyclically looks for the optimal pitch angle according to the evolution of the rotational speed. This evolution of the pitch angle is shown in diagram **420** in revolutions per minute (rpm).
(34) Diagram **410** shows the pitch angle in degrees. Changes in the pitch angle originate from the calculated pitch angle adjustments.
(35) Diagram **430** shows the cost in rpm which corresponds to the incremental difference between the rotational speed before and after the pitch angle change, this change due to a calculated and applied pitch angle adjustment.
(36) In this example, the cost function starts with a pitch angle adjustment towards feather. This calculated pitch angle adjustment is applied as a pitch step towards feather. Feathering means that the blade pitch angle is increased to the point that the chord line of the blade is approximately parallel to the on-coming airflow. This approach minimizes the unwanted effect of stalling the wind turbine.
(37) In diagram **410**, this first pitch angle adjustment is depicted as the first increase of the pitch angle. As a consequence, the rotational speed decreases as is shown in diagram **420**. This decrease in speed is unwanted. Hence, the cost decreases as well as is depicted in diagram **430**.
(38) As described, the first pitch angle adjustment towards feather does not lead to an increase of the speed of the generator **140** and thus decreases cost. As the cost function aims to maximize cost a pitch angle adjustment towards fine i.e., to the opposite is introduced.
(39) As the rotational speed as well as the cost decreases further a further step towards fine having a smaller amplitude is introduced. As a consequence, the rotational speed increases slowly over time.
(40) Hence, the cost function has found a way to maximize the generator speed and issues some increasing pitch angle adjustments with the result of increasing rotational speed and cost.
(41) The amplitudes of the pitch angle adjustment and pitch steps may be adaptive with respect to the number of search iterations, as the optimal pitch angle should be closer with the increasing number of iterations.
(42) As a termination condition of the iteration, the method may be terminated after sufficient rotational speed is measured to connect the generator **140** to the grid, a predetermined time limit is reached without being able to start-up the wind turbine **100** and/or no ice is detected anymore.
(43) Although the present invention has been disclosed in the form of embodiments and variations thereon, it will be understood that numerous additional modifications and variations could be made thereto without departing from the scope of the invention.
(44) For the sake of clarity, it is to be understood that the use of “a” or “an” throughout this application does not exclude a plurality, and “comprising” does not exclude other steps or elements.

Claims

1. A method for starting up a wind turbine for generating electrical power, wherein ice accretion is on at least one blade of a rotor of the wind turbine and wherein the at least one blade has a variable pitch angle, the method comprising: measuring a rotational speed of an electromechanical transducer; calculating a pitch angle adjustment with a goal of maximizing a rotational speed of the

wind turbine; modifying the variable pitch angle in a positive or negative step according to the pitch angle adjustment; and iterate the preceding steps, whereby the method cyclically looks for an optimal pitch angle according to an evolution of the rotational speed, wherein the pitch angle adjustment is calculated with a cost function, the cost function receiving the rotational speed of the electromechanical transducer as an input with the goal of maximizing the rotational speed.

2. The method as set forth in claim 1, wherein as a first step, ice accretion is detected.

3. The method as set forth in claim 1, wherein as a last step, the method is terminated after sufficient rotational speed is measured to connect the electromechanical transducer to a grid, a predetermined time limit is reached without being able to start-up the wind turbine and/or no ice is detected.

4. The method as set forth in claim 1, wherein the measured rotational speed of the electromechanical transducer is filtered with a low pass and/or a band pass.

5. The method as set forth in claim 1, wherein a first pitch angle adjustment is a pitch step towards feather.

6. The method as set forth in claim 5, wherein if the pitch angle adjustment towards feather does not lead to an increase of the speed of the electromechanical transducer a pitch angle adjustment towards fine is introduced.

7. The method as set forth in claim 1, wherein amplitudes of the pitch angle adjustments are adaptive with respect to a number of iterations.

8. A wind turbine control system for controlling a wind turbine having an electromechanical transducer, the control system comprising: a computation unit configured for executing the method as set forth in claim 1; an input configured for receiving a rotational speed of an electromechanical transducer; and an output configured for outputting a pitch angle adjustment.

9. The wind turbine control system as set forth in claim 8, wherein the computation unit comprises a low pass and/or a band pass filter configured for filtering the measured rotational speed of the electromechanical transducer.

10. A wind turbine for generating electrical power, the wind turbine comprising: a tower; a rotor, which is arranged at a top portion of the tower and which comprises at least one blade, wherein the at least one blade has a variable pitch angle; an electromechanical transducer which is mechanically coupled with the rotor; and a wind turbine control system as set forth in claim 8.

11. The wind turbine as set forth in claim 10, wherein the wind turbine comprises a rotational speed sensor configured for measuring the rotational speed of the electromechanical transducer and/or a pitch sensor configured for measuring a pitch angle of the at least one blade.